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Группа: РПО-3 2025

Дисциплина: Компьютерные сети

Вид работы: Домашняя работа

```
!
!
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface FastEthernet0/0.2
 encapsulation dot1Q 2
 ip address 192.168.2.1 255.255.255.0
!
interface FastEthernet0/0.3
 encapsulation dot1Q 3
 ip address 192.168.3.1 255.255.255.0
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
```

Рис. 1

Настройка сабинтерфейсов VLAN.

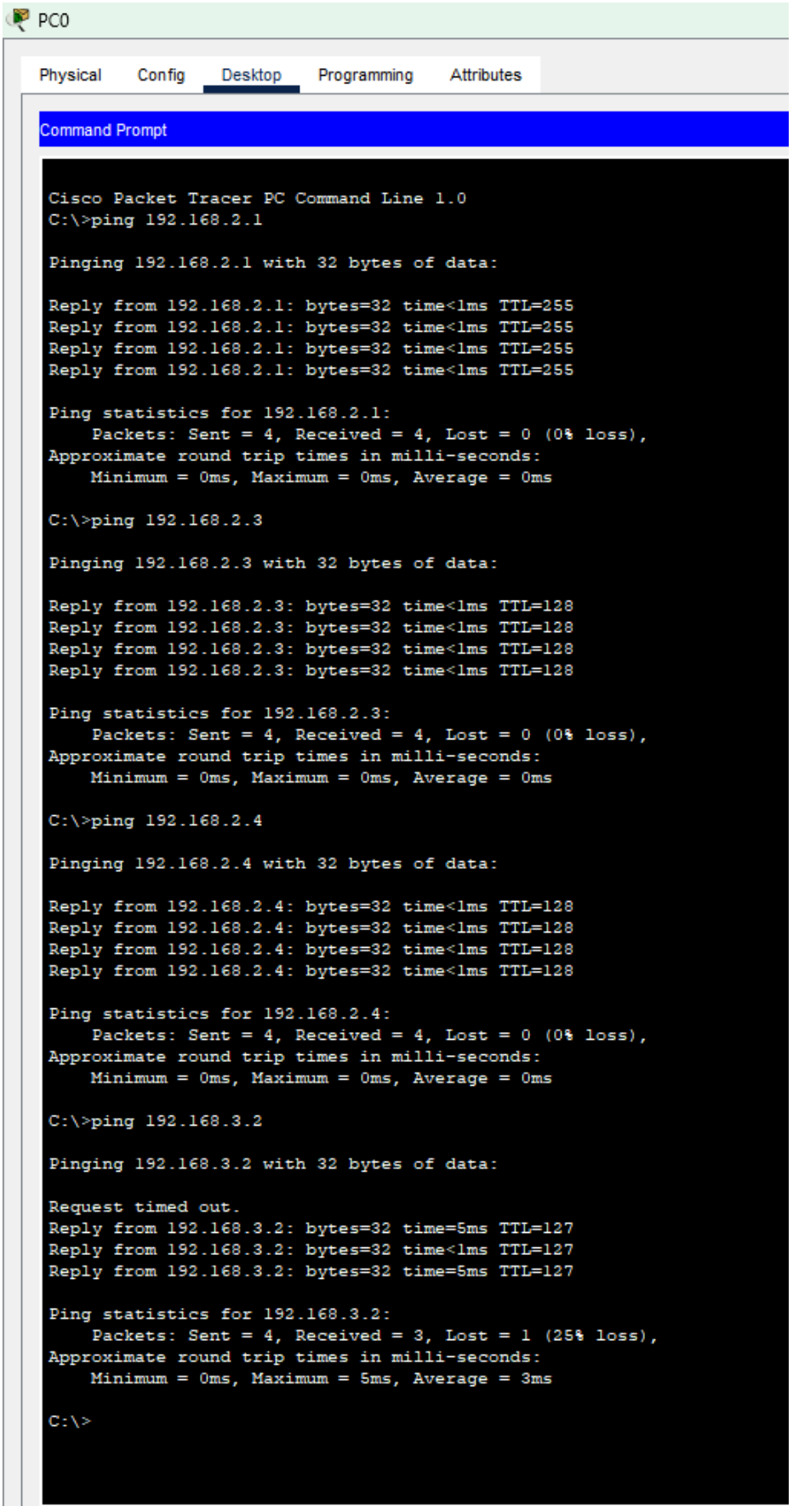

PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1
Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.2.4
Pinging 192.168.2.4 with 32 bytes of data:
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.2.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.3.2
Pinging 192.168.3.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.3.2: bytes=32 time=5ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time=5ms TTL=127
Ping statistics for 192.168.3.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 3ms
C:\>

Рис. 2

Проверка связи внутри VLAN.

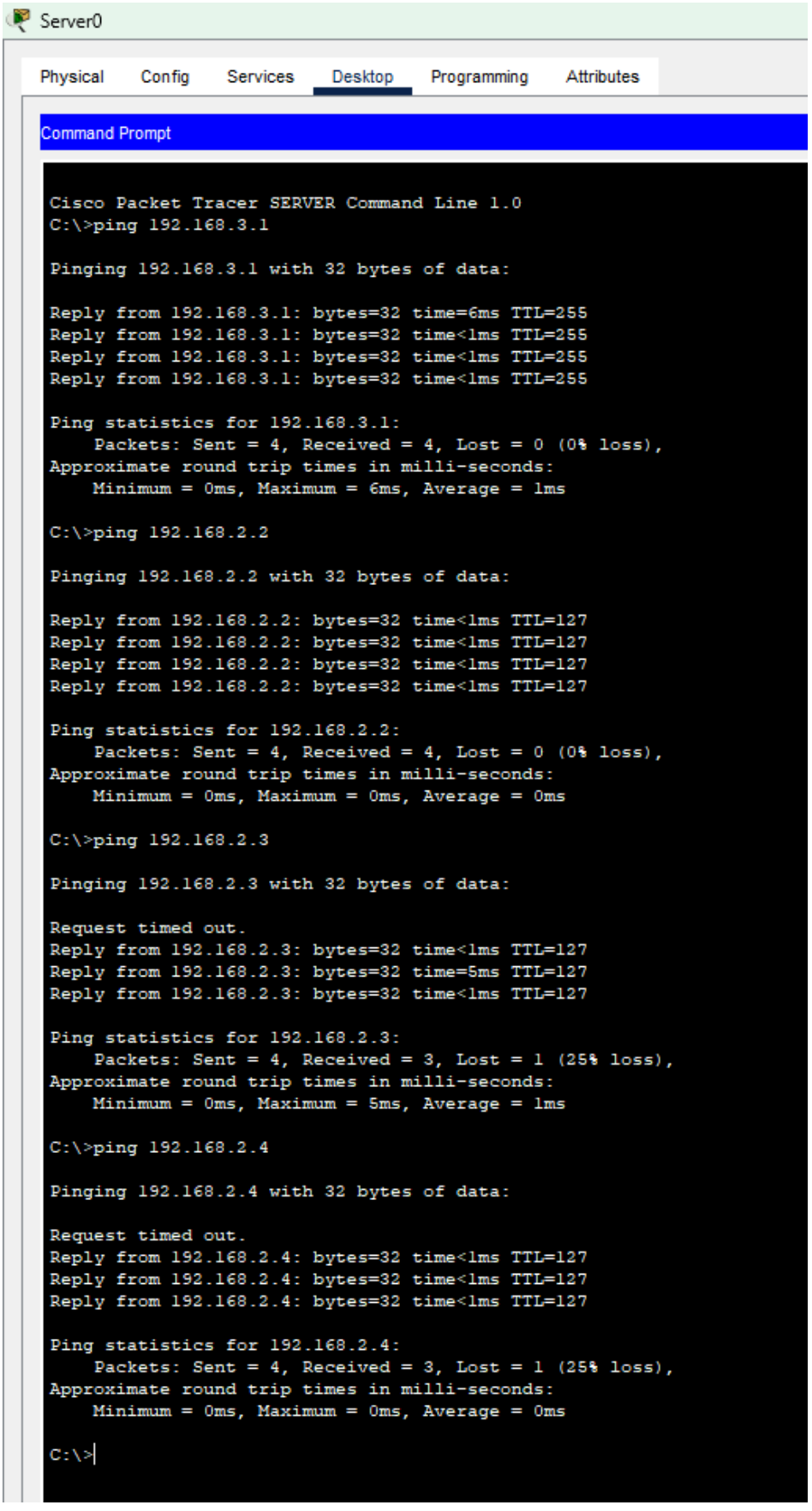


Рис. 3

Связь сервера с ПК и шлюзом.

```
!
interface FastEthernet0/0
 ip address 213.234.10.1 255.255.255.252
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 213.234.20.1 255.255.255.252
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
 --More--
```

Рис. 4

Настройка интерфейсов провайдера.

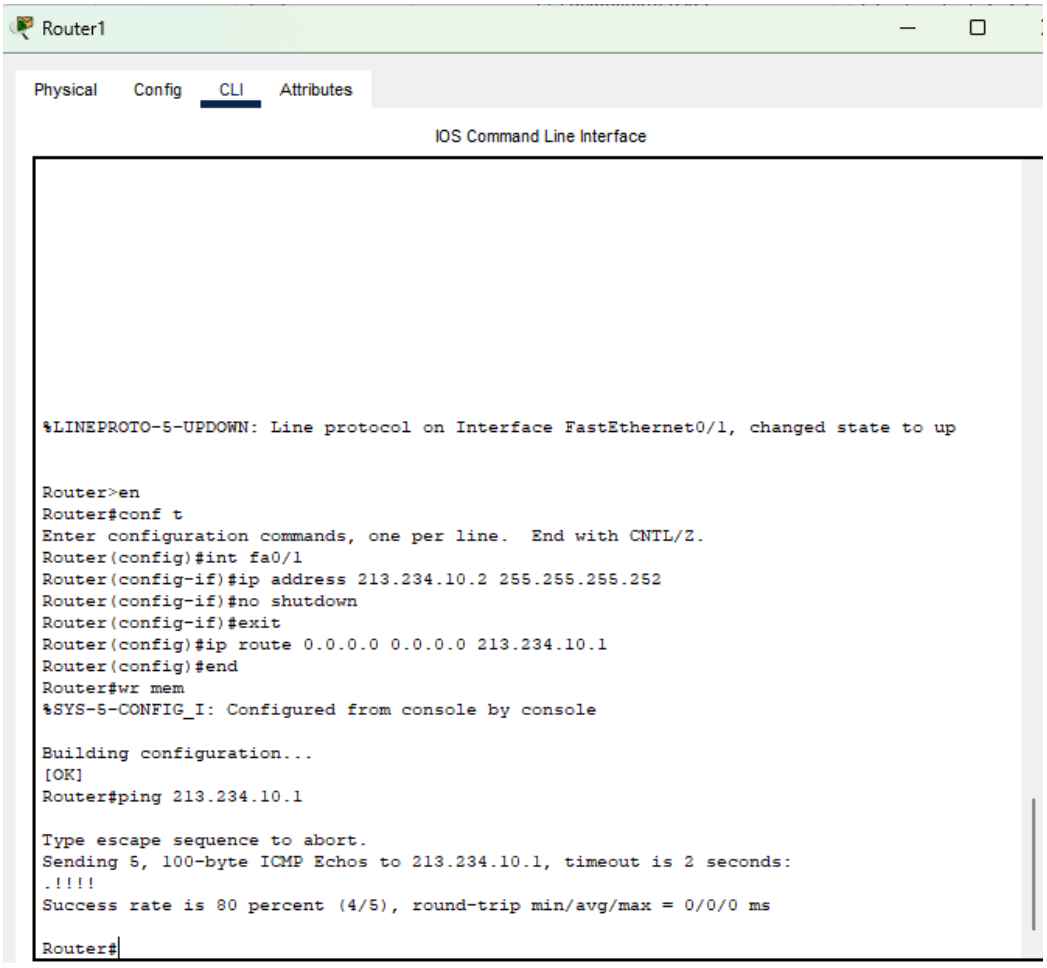


Рис. 5

Настройка внешнего интерфейса и шлюза.

```
C:\>ping 213.234.20.2

Pinging 213.234.20.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 213.234.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Рис. 6

Отсутствие связи до настройки NAT.

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
!
!
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface FastEthernet0/0.2
 encapsulation dot1Q 2
 ip address 192.168.2.1 255.255.255.0
 ip nat inside
!
interface FastEthernet0/0.3
 encapsulation dot1Q 3
 ip address 192.168.3.1 255.255.255.0
 ip nat inside
!
interface FastEthernet0/1
 ip address 213.234.10.2 255.255.255.252
 ip nat outside
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 0.0.0.0 0.0.0.0 213.234.10.1
!
ip flow-export version 9
!
!
ip access-list standard FOR-NAT
 permit 192.168.2.0 0.0.0.255
 permit 192.168.3.0 0.0.0.255
!
```

Copy Paste

Рис. 7

Конфигурация NAT и ACL.

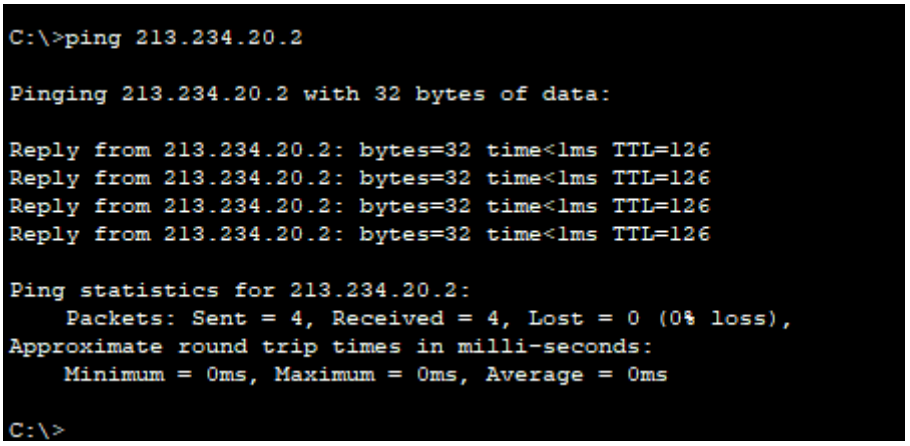


Рис. 8
Работа PAT (доступ в Интернет).

```
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
tcp 213.234.10.2:80    192.168.3.2:80     ---                ---

Router#
```

Рис. 9
Таблица трансляций Static NAT.

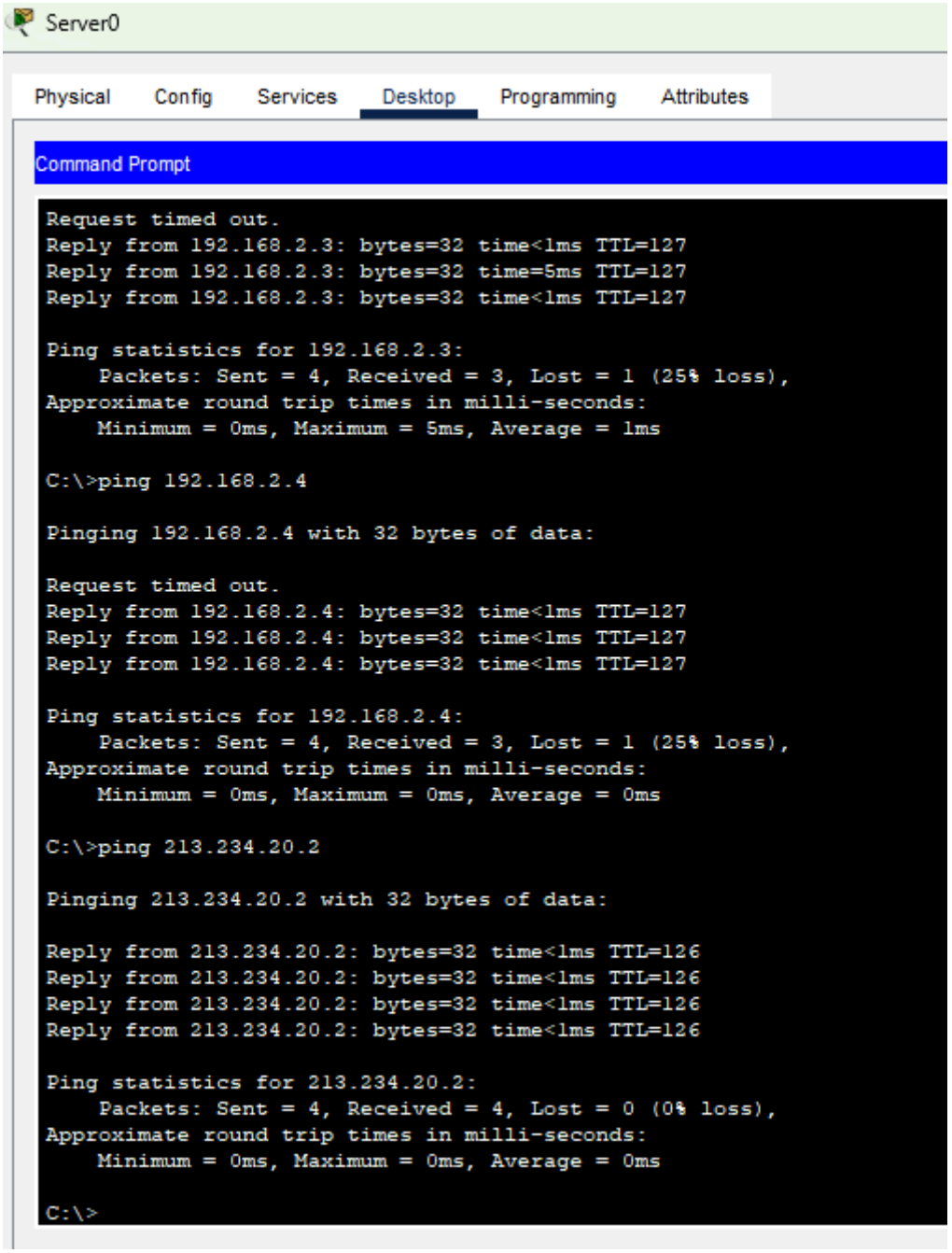


Рис. 10

Связь сервера с внешней сетью.

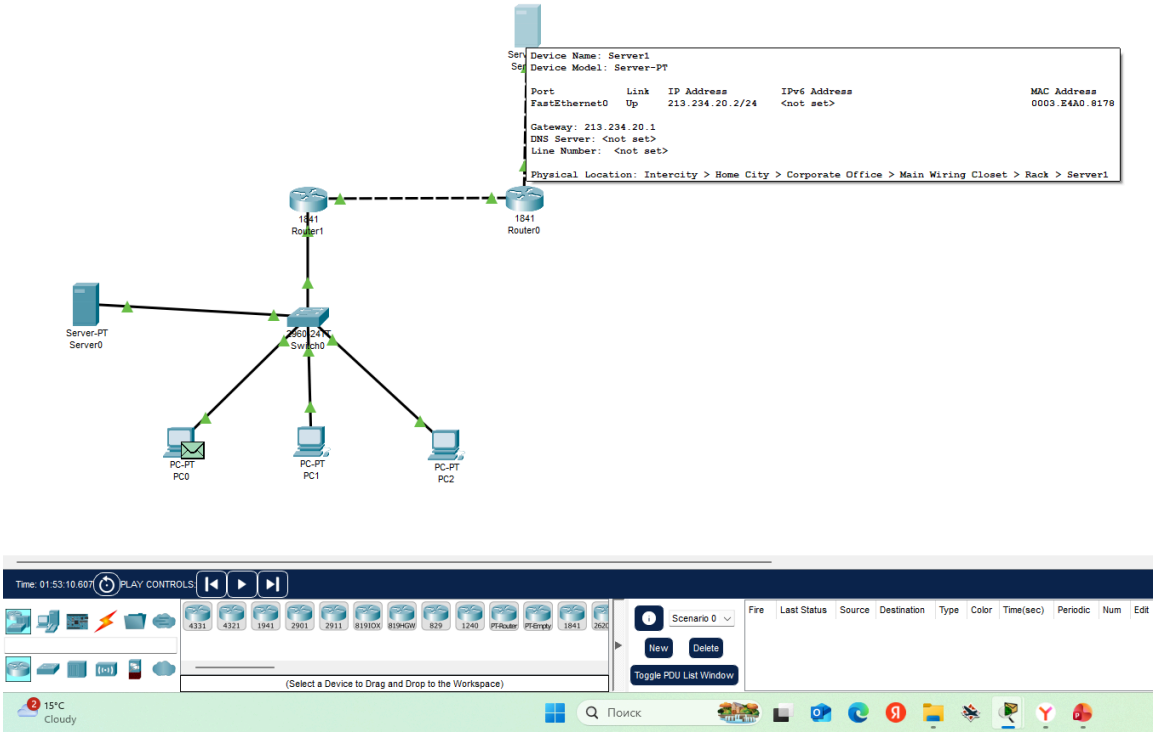


Рис. 11

Параметры внешнего сервера.

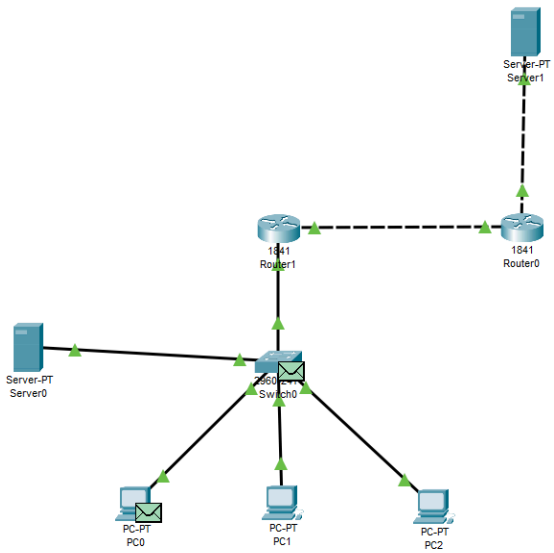


Рис. 12

Доход письма до роутера.

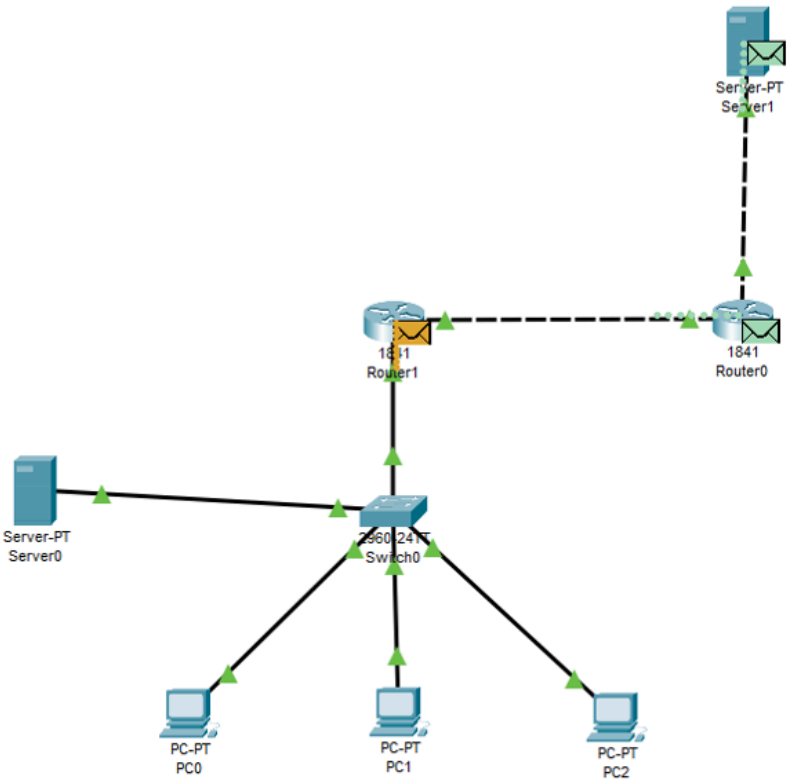


Рис. 13
Схема прохождения почты.

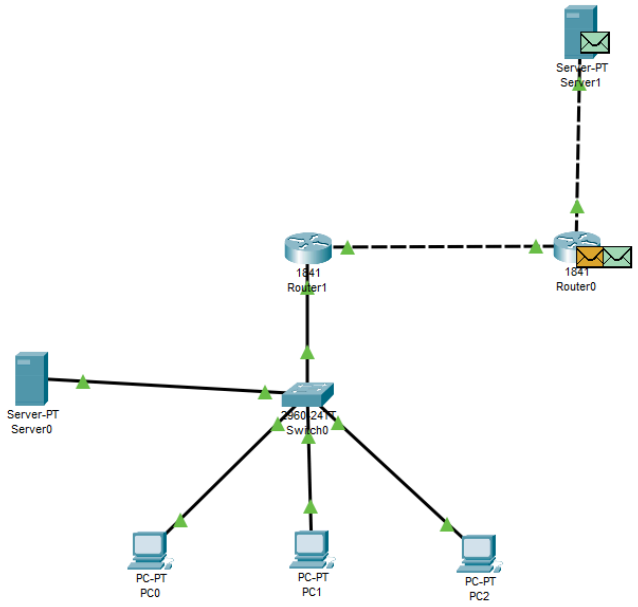


Рис. 14
Доставка почты через Router 0.

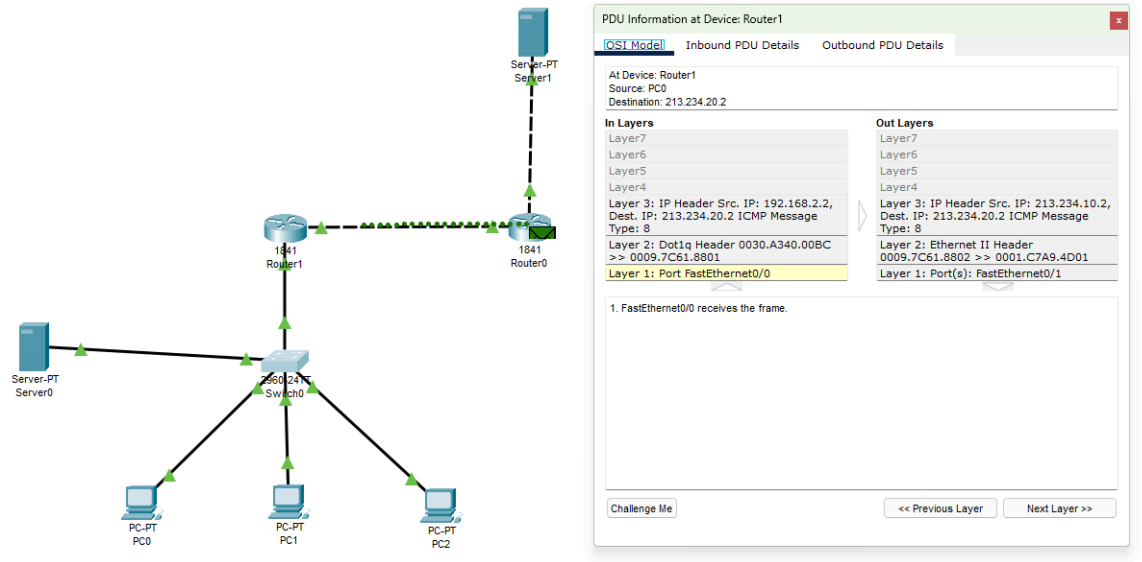


Рис. 15

Трансляция IP-адреса (PAT).

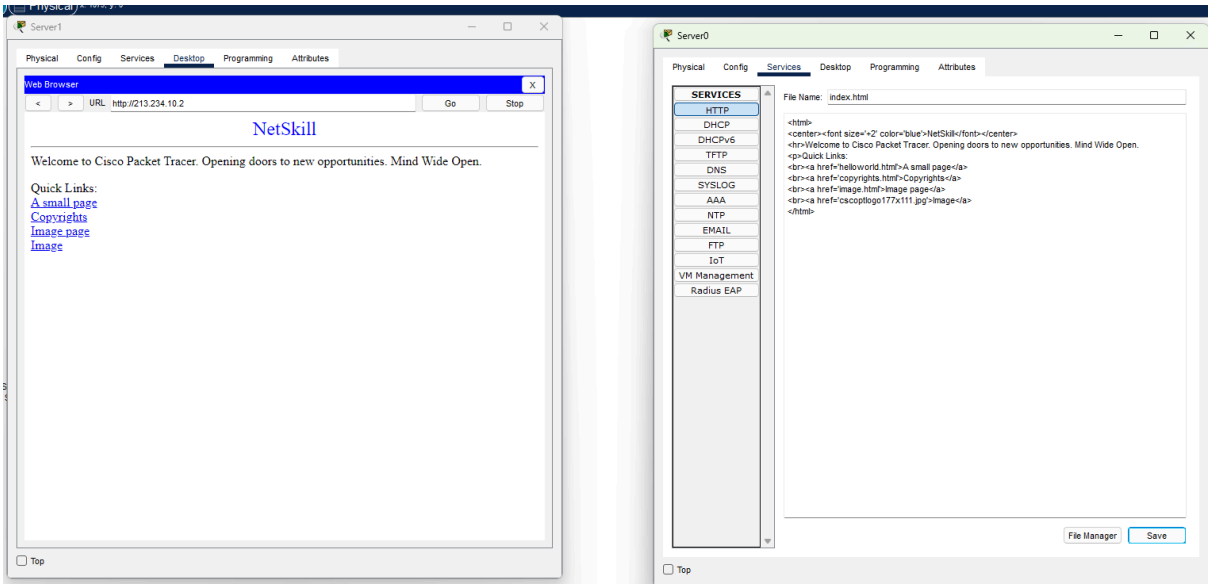


Рис. 16

Работа Static NAT (доступ к веб-серверу).